

Statistik_2023-01-30_D

1)

1. In a linear regression model (y_i modeled as a linear function of x_i plus error') the parameters are estimated via least squares. For the mean and the empirical standard deviation of the x and y values we obtain $\bar{x} = 3$, $s_x = 4$, $\bar{y} = 7$ and $s_y = 3$. It holds that
- (a) the slope of the regression line is smaller than $-3/4$
 - (b) the regression line goes through $(3, 7)$**
 - (c) the regression line goes through $(7, 3)$
 - (d) the slope of the regression line is larger than $3/4$

It always goes through $(\bar{x}, \bar{y})$

2)

2. Which one of the following is an **incorrect** statement?

- (a) The larger the value of the sample size n , the closer the standard deviation of the sampling distribution of $\bar{x}$ is to the standard deviation of the population.
- (b) The sampling distribution of $\bar{x}$ has mean equal to the population mean μ even if the population is not normally distributed.
- (c) When n is large, the sampling distribution of $\bar{x}$ is approximately normal even if the population is not normally distributed.
- (d) The sampling distribution of $\bar{x}$ has standard deviation $\sigma/\sqrt{n}$ even if the population is not normally distributed.

Standard deviation of the population is σ while the SD of the sampling distribution is $\frac{\sigma}{\sqrt{n}}$. If n gets bigger, they don't get closer.

Why the others are correct

1. **Mean equals population mean:** The mean of the sampling distribution is always the same as the population mean ($\mu_{\bar{x}} = \mu$). This is true for any distribution.
2. **Normal approximation (CLT):** The **Central Limit Theorem** states that if n is large enough (usually $n \geq 30$), the sampling distribution becomes approximately normal, even if the original population is weird or skewed.
3. **Formula holds:** The formula $StandardError = \frac{\sigma}{\sqrt{n}}$ works regardless of the population's shape, as long as the samples are independent.

3)

3. Two features of a novel operating system are compared using a two-sample t -test. The statistics for the first feature are $\bar{x} = 21$, $s_x = 10$ and $n_x = 4$ and those for the second features are $\bar{y} = 29$, $s_y^2 = 55$ and $n_y = 5$. The rejection region is given through $R = (-\infty, -q] \cup [q, \infty)$. Then it holds

- (a) we reject for $q = 0.4$ but not for $q = 1.2$
- (b) we do not reject for $q = 0.4$ but for $q = 1.2$
- (c) we reject for both $q = 0.4$ and $q = 1.2$
- (d) we do neither reject for $q = 0.4$ nor for $q = 1.2$

$$H_0 : \mu_x - \mu_y = 0$$

$$T = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{\sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}}}$$

$$T = \frac{21 - 29 - 0}{\sqrt{\frac{100}{4} + \frac{55}{5}}} = \frac{-8}{6} \approx -1.33$$

4)

4. Let X be a random variable with a Poisson distribution. If it holds

$$P(X = 1) = P(X = 3),$$

then the expectation $\mathbb{E}X$ equals

- (a) 6
- (b) 3
- (c) $\sqrt{6}$
- (d) $\sqrt{3}$

$$f(x) = e^{-\lambda} \frac{\lambda^x}{x!} \implies P(X = 1) = P(X = 3) \implies \frac{\lambda^1}{1!} e^{-\lambda} = \frac{\lambda^3}{3!} e^{-\lambda}$$

$$\lambda = \frac{\lambda^3}{6}$$

$$6\lambda = \lambda^3$$

$$\lambda^3 - 6\lambda = 0$$

$$\lambda(\lambda^2 - 6) = 0$$

- $\lambda_1 = 0$
- $\lambda^2 - 6 = 0 \Rightarrow \lambda^2 = 6 \Rightarrow \lambda = \sqrt{6}$

5)

5. In general, how does halving the sample size change the confidence interval size?

- (a) Doubles the interval size
 - (b) Halves the interval size
 - (c) Divides the interval size by $\sqrt{2}$
 - (d) Multiplies the interval size by $\sqrt{2}$
-

Width of a CI:

$$W = 2 \cdot z \cdot \frac{\sigma}{\sqrt{n}}$$

6)

6. A fast food chain advertises that their large bag of french fries has a weight of 150 grams. Some high school students, who enjoy french fries at every lunch, suspect that they are getting less than the advertised amount. With a scale borrowed from their physics teacher, they weigh a random sample of 16 bags. Assuming the level of significance $\alpha = 10\%$, what would be the conclusion if the sample mean is 144 g and standard deviation is 15 g? Assume that all conditions for inference are met.

- (a) There is sufficient evidence to prove the fast food chain advertisement is true.
- (b) The students do not have sufficient evidence to reject the fast food chain's claim.
- ☒ (c) The students have sufficient evidence to reject the fast food chain's claim.
- (d) There is sufficient evidence to prove the fast food chain advertisement is false.

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

$$t = \frac{144 - 150}{\frac{15}{\sqrt{16}}} = \frac{-6}{\frac{15}{4}} = \frac{-6}{3.75} = -1.6$$

t-Table:

$$significance = \alpha = 0.1$$

$$df = n - 1 = 15$$

$$\Rightarrow 1.341$$

left sided test $\Rightarrow -1.341 \Rightarrow -1.6 < -1.341 \Rightarrow$ rejection possible

7)

7. We toss two fair coins simultaneously and independently. If the outcomes of the two coin tosses are the same, we win, otherwise, we lose. Let A be the event that the first coin comes up heads, B be the event that the second coin comes up heads and C be the event that we win. Which one of the following statements is **true**?
- (a) The probability of winning is $3/4$.
 - ☒ (b) Events A and C are independent.
 - (c) Events A and B are not independent.
 - (d) Events B and C are not independent.

8)

8. Which of the following statements about t -istribution are **true**?

- I Like the normal, t -distributions are always symmetric.
 - II The smaller the number of degrees of freedom, the closer the curve is to the normal curve.
 - III Twenty degrees of freedom gives the normal curve.
- (a) I and II
 - (b) III only
 - (c) I only
 - (d) I and III
-

i) correct

ii) wrong, its the other way around

iii) not identical... (u only start to use z-test if you have $n \geq 30$)

9)

9. The income per household in a certain country is assumed to be normally distributed with the mean 9500 Euro and standard deviation of 1750 Euro. The middle 95% of incomes (in Euro) are between what two values?

- (a) 5422 and 13578
- (b) 6621 and 12379
- ☒ (c) 6070 and 12930
- (d) 8049 and 10951

$$CI = \bar{x} \pm z \left(1 - \frac{\alpha}{2} \right) \cdot \sigma$$

$$CI = 9500 \pm z \left(1 - \frac{\alpha = 1 - 0.95 = 0.05}{2} \right) \cdot 1750 = 9500 \pm z(0.9750) \cdot 1750$$

$$CI = 9500 \pm 1.96 \cdot 1750 = 12930 \text{ and } 6070$$

10)

10. Let $X \sim Poi(2)$ and $Y \sim Bin(8, 0.5)$ be two random variables with the correlation $Corr(X, Y) = 0.4$. Compute $Cov(X, Y + 3)$.

- (a) 1.6
- (b) 0.8
- (c) -2.2
- (d) -1.4

Schritt 1: Rechenregel für Kovarianz anwenden

Konstanten ändern die Kovarianz nicht (Verschiebungsinvarianz).

$$Cov(X, Y + 3) = Cov(X, Y)$$

Schritt 2: Beziehung zwischen Kovarianz und Korrelation

$$Cov(X, Y) = Corr(X, Y) \cdot \sigma_X \cdot \sigma_Y$$

Schritt 3: Standardabweichungen (σ) berechnen

- Für Poisson (X): $Var(X) = \lambda = 2$

$$\sigma_X = \sqrt{2}$$

- Für Binomial (Y): $Var(Y) = n \cdot p \cdot (1 - p) = 8 \cdot 0.5 \cdot 0.5 = 2$

$$\sigma_Y = \sqrt{2}$$

Schritt 4: Einsetzen

$$\begin{aligned} Cov(X, Y) &= 0.4 \cdot \sqrt{2} \cdot \sqrt{2} \\ &= 0.4 \cdot 2 \\ &= 0.8 \end{aligned}$$

11)

11. Let $X_1, \dots, X_{64}$ be a random sample from a distribution with the expectation -1.2 and variance 4. Let

$$\bar{X} = \frac{1}{64} \sum_{i=1}^{64} X_i$$

be the sample mean. Determine the approximate value of $P(\bar{X} > -0.9)$ using the Central limit theorem and express it in terms of a suitable R-function.

- (a) `pnorm(-0.9, -1.2, 0.25)`
- (b) `pnorm(1.2)`
- (c) `pnorm(-1.2)`
- (d) `pnorm(-0.9, 1.2, 0.5)`

1. Standardfehler berechnen

Da wir den Mittelwert einer Stichprobe betrachten, verringert sich die Streuung:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{64}} = \frac{2}{8} = 0.25$$

2. z-Transformation (Standardisierung)

Wir rechnen den Grenzwert -0.9 in einen z-Wert um:

$$z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} = \frac{-0.9 - (-1.2)}{0.25} = \frac{0.3}{0.25} = 1.2$$

Das Problem reduziert sich also auf: **Gesucht ist $P(Z > 1.2)$.**

3. Auswahl des R-Befehls

- Der Befehl `pnorm(z)` berechnet die kumulative Verteilungsfunktion $P(Z \leq z)$ (Fläche nach links).
- Wir suchen die Fläche nach rechts (größer als 1.2).
- Symmetrie-Eigenschaft:** Die Fläche rechts von 1.2 ist identisch mit der Fläche links von -1.2 .

$$P(Z > 1.2) = P(Z < -1.2)$$

12)

12. Suppose the null hypothesis $H_0 : p = 0.4$, and the power of the test for the alternative hypothesis $H_A : p = 0.35$ is 0.75. Which of the following is a valid conclusion?
- (a) If the null hypothesis is false, the probability of failing to reject it is 0.6
 - (b) The probability of committing a Type I error is 0.05.
 - (c) The probability of committing a Type II error is 0.65.
 - (d) If the alternative hypothesis is true, the probability of failing to reject the null is hypothesis 0.25.
-

$$1 - 0.75 = 0.25$$

13)

13. Two classes take the same exam. Suppose a certain score is at the 40th percentile for the first class and at the 80th percentile for the second class. Which of the following is the most reasonable conclusion?
- (a) One of the classes has twice the number of students as the other.
 - (b) Students in the second class generally scored higher than students in the first class.
 - ☒ (c) Students in the first class generally scored higher than students in the second class.
 - (d) A score at the 50th percentile for the first class is at the 90th percentile for the second class.
-

class A: 40% of students were worse or as good as X

class B: 80 % of students were worse or as good as X

That means that 60% of class A were better than X while only 20% of students in class B performed at that rate...

14)

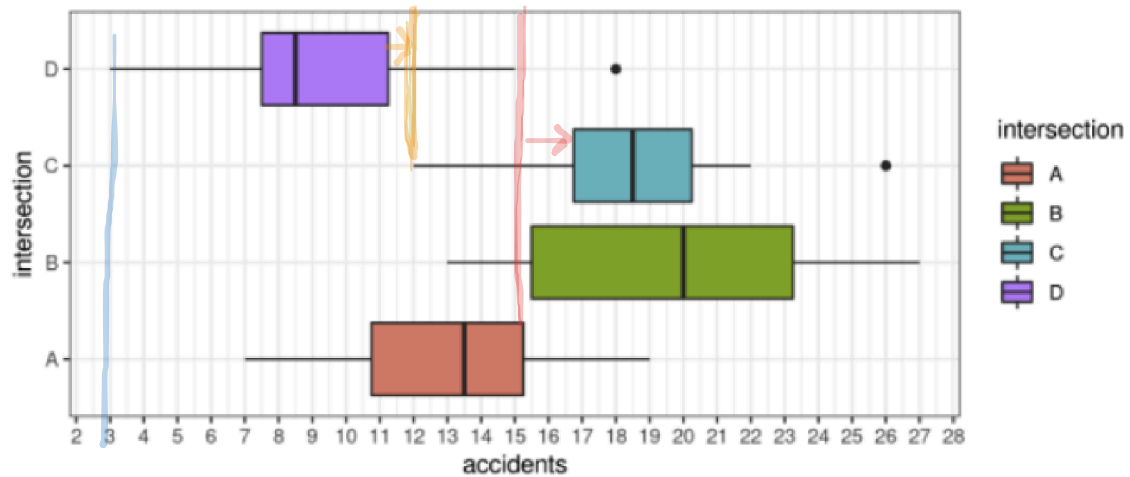
14. Out of the students in a class, 60% are playing chess, 70% love ice skating, and 40% fall into both categories. Compute the probability that a randomly selected student is neither a chess player nor an ice skating lover.

- (a) 0.4
- (b) 0.1**
- (c) 0.6
- (d) 0.9

$$P(\text{neither}) = 1 - (0.6 + 0.7 - 0.4) = 0.1$$

15)

15. Data on the number of yearly accidents were collected from four intersections (A-D) over a 20 year period and are presented below. Which one of the following statements is **false**?



- (a) During at least 5 years, fewer than 10 accidents occurred at section A. ✗
- (b) During at least 75% of years, intersection C had more accidents than the lowest 75% of years at intersection A. ✓
- (c) The minimum accident total at intersection C was higher than the number of accidents observed at intersection D in 75% of years. ✓
- (d) The minimum number accidents that occurred in a single intersection was 3. ✓

16)

16. For a project, a high school student randomly picks 100 fellow Statistics students to survey on whether each has either a PC or Apple at home (all students in the school have a home computer) and what score (1, 2, 3, 4, 5) each expects to receive on the Statistics exam. A chi square test of independence results in a test statistic of 8. How many degrees of freedom are there?

- (a) 4
- (b) 9
- (c) 7
- (d) 1

$$df = (\text{number of scores} - 1) \cdot (\text{number of computer types} - 1) = 4 \cdot 1$$

17)

17. For a statistical test of significance level α it holds

- (a) the rejection area does not depend α
- (b) rejection at level α implies rejection at level $\alpha/2$
- (c) the rejection area shrinks when α is increased
- (d) the rejection area depends on the distribution of the test statistic under the null hypothesis

-
- **Der "Ablehnungsbereich"** besteht aus den konkreten Werten auf der x-Achse, die dazu führen, dass wir H_0 verwerfen (z. B. "alles größer als 1,96").
 - **Abhängigkeit von der Verteilung:** Da eine t-Verteilung beispielsweise breiter ist als eine Normalverteilung, liegen die Grenzwerte (kritischen Werte) woanders, um dieselben 5% Fläche abzuschneiden. Deshalb hängt der Bereich tatsächlich von der Form der Kurve (der Verteilung) ab.

18)

18. A study is to be performed to estimate the proportion of voters who believe the economy is "heading in the right direction." Which of the following pairs of sample size n and population proportion p will result in the smallest variance for the sampling distribution of $\hat{p}$?

- (a) $n = 100$ and $p = 0.1$
- (b) $n = 100$ and $p = 0.99$
- (c) $n = 1000$ and $p = 0.5$
- (d) $n = 1000$ and $p = 0.1$

$$\text{Var}(\hat{p}) = \frac{p \cdot (1 - p)}{n}$$

$\Rightarrow n$ as big as possible

$$\text{Var}(\hat{p}) = \frac{0.5 \cdot (1 - 0.5)}{1000} = 0.00025$$

$$\text{Var}(\hat{p}) = \frac{0.1 \cdot (1 - 0.1)}{1000} = 0.00009$$

19)

19. Consider the two sets $X = \{10, 30, 45, 50, 55, 70, 90\}$ and $Y = \{10, 30, 35, 50, 65, 70, 90\}$. Which one of the following answers is **false**?

- (a) The sets have identical ranges.
 - (b) The sets have identical medians.
 - ☒ (c) None of the rest are false.
 - (d) The sets have identical means.
-

a) 10 - 90 -> true

b) 50 and 50 -> true

c) all of the other are correct -> true...

d) $\text{mean}(X) = 50$, $\text{mean}(Y) = 50$

i guess they want me to cross c...

20)

20. Let $X \sim \mathcal{N}(-1, 4)$. Express the probability

$$P(X^2 + 2X \leq 0)$$

in terms of the cumulative distribution function Φ of the standard normal random variable

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt, \quad x \in \mathbb{R}.$$

- (a) $\Phi(2) - 0.5$
- (b) $2 \cdot \Phi(0.25) - 1$
- (c) $2 \cdot \Phi(0.5) - 1$
- (d) $2 - 2 \cdot \Phi(0.5)$

$$P(X^2 + 2X \leq 0) \implies P(X(X+2)) \implies P(-2 \leq X \leq 0)$$

$X \sim \mathcal{N}(-1, 4)$.

- $\mu = -1$.
- variance = 4 $\implies$ std $\sigma = \sqrt{4} = 2$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

With our values:

$$z_{-2} = \frac{-2 - (-1)}{2} = -\frac{1}{2} \quad z_0 = \frac{0 - (-1)}{2} = \frac{1}{2}$$

We search for:

$$P(-0.5 \leq Z \leq 0.5) = \Phi(0.5) - \Phi(-0.5)$$

We know that $1 - \Phi(0.5) = \Phi(-0.5)$

$$\begin{aligned} \implies & \Phi(0.5) - (1 - \Phi(0.5)) \\ & \Phi(0.5) + \Phi(0.5) - 1 \\ & 2\Phi(0.5) - 1 \end{aligned}$$